

13 Abridged: Introduction to Catalysis

A catalyst is a reactive intermediate that promotes production of a large number of product molecules

Catalysts provide lower energy routes from reactants to products without being consumed.

The catalyst influences kinetics, but they do not influence the thermodynamics of the overall reaction

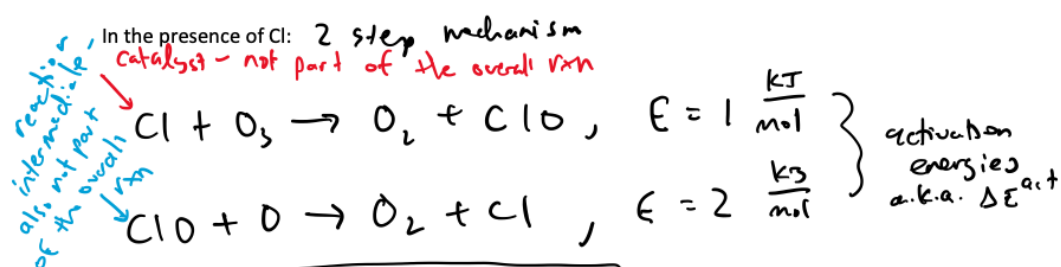
Types of catalysts:

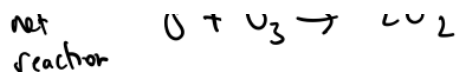
- Homogeneous catalysts: reaction fluids & catalyst are in the same phase

Examples: Ion/acid catalysts, organometallic catalysts, enzymes

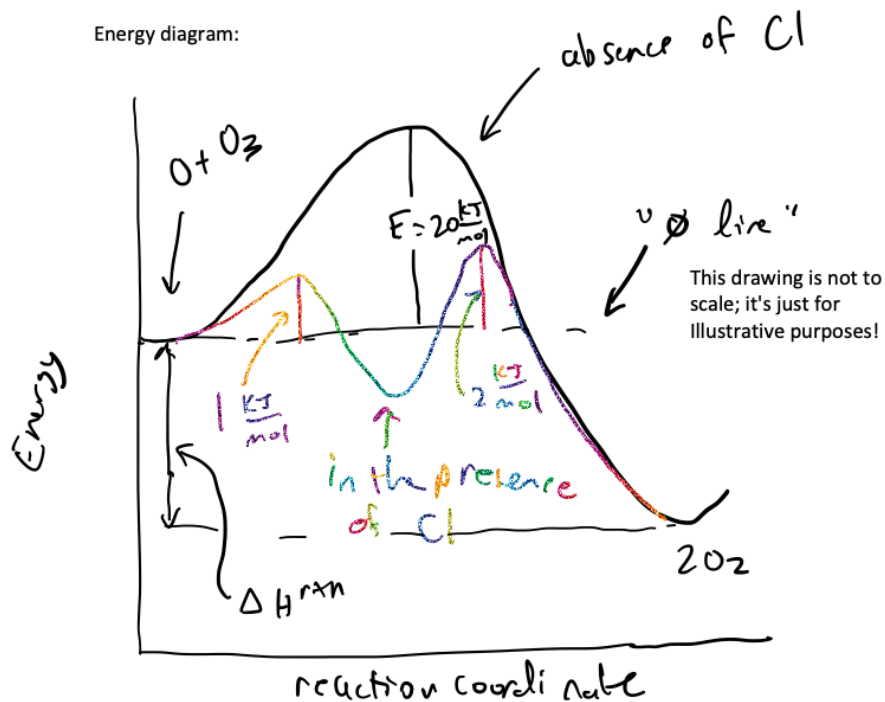
Example: Cl atoms catalyze the decomposition of ozone O_3 :

In the absence of Cl: $O + O_3 \rightarrow 2O_2$, activation energy $E = 20 \text{ kJ/mol}$





"same reaction, but kinetic pathway is more favourable"



The catalyst provides a lower energy pathway from reactants to products. Alters the kinetics. DOES NOT alter the thermodynamics $\rightarrow \Delta H_{rxn}$, ΔG_{rxn} , etc $\rightarrow$ These are

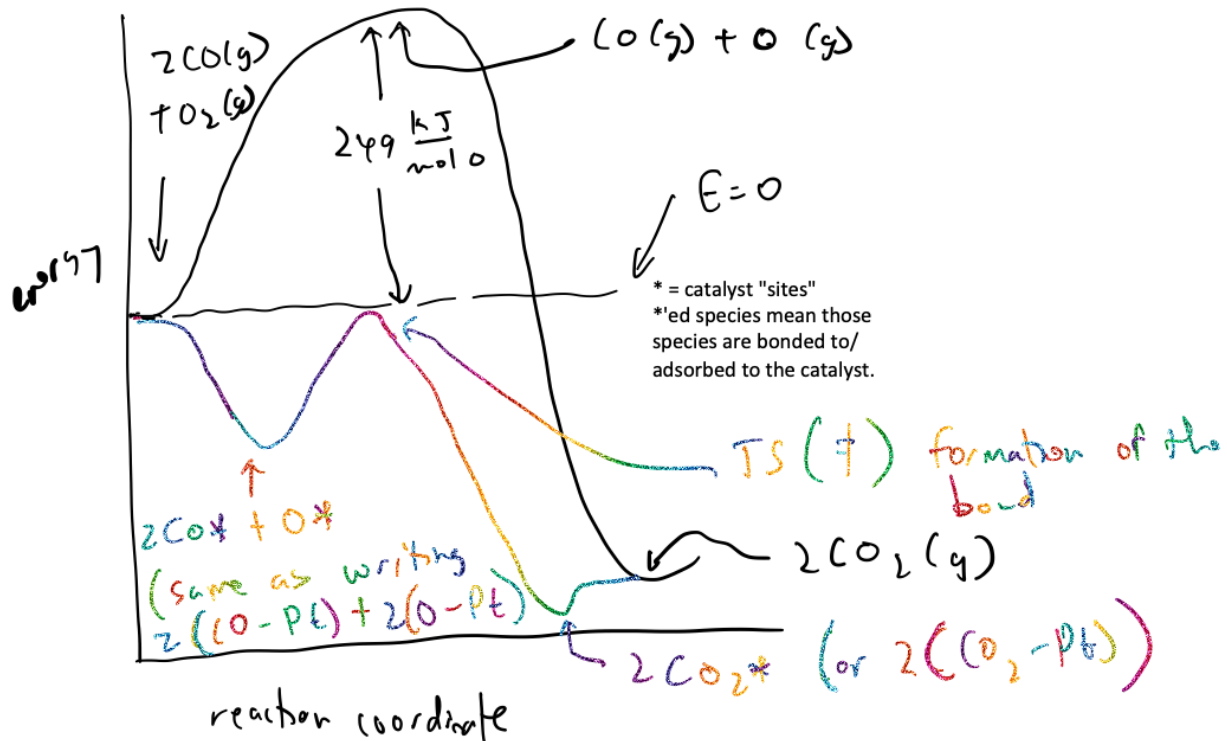
STATE FUNCTIONS -

- Heterogeneous catalysts:

different phase than reaction fluids—typically solid whereas reaction fluids would be liquid or gas

Examples: Supported metal catalysts, zeolite (molecular sieves), Ziegler-Natta catalyst (TiCl_4 anchored to MgCl_2)

Example: CO oxidation on $\text{Pt}/\text{Al}_2\text{O}_3$ $2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g})$



O_2 doesn't dissociate in the gas phase, but it readily adsorbs to the catalyst and dissociates on the catalyst. The catalyst provides a lower energy route from reactants to products.

